MindOrigin / Noah — P/G Behavioral Layer

MVP Executable Specification (v1.6.1)

Status: schema/contract-frozen; scalar aggregation functions deferred to calibration · June 2026 · Self-contained; supersedes v1–v1.5 and the unified design record

How to read this document

This is a single, self-describing document. It merges the conceptual foundation of the P/G framework with the frozen implementation specification of its behavioral layer, and integrates every fix from the most recent external review. You do not need any prior version to read it. Part I explains what we are building and why. Part II defines the representations (P, G, relations). Part III is the implementation specification: data model, schemas, modules, and rules. Part IV is validation: the benchmark suite, grading, and acceptance test. Part V records what is deferred and the build order.

Passages marked in green are additions made for v1.6 in response to the latest review — operational rules (dimension assignment, adapter contract, metric definitions, the member_welfare form) rather than new philosophy. v1.6.1 adds a consistency layer on top: it renders the FACT_TYPE vocabulary, fixes the member_welfare example, defines delta saturation, separates magnitude from uncertainty bands, and clarifies M4/M5 ownership — no new theory. The functional forms are frozen; their value-laden constants (λ, w_P) are deferred to the ΔG calibration session, so this is a contract-frozen executable spec, not a fully sealed one. Code blocks are normative: where prose and a schema disagree, the schema governs.

Part I — Foundations: what we are building and why

1. The problem

Before acting, people instinctively estimate the social consequences of an action: who is affected, in what way, how much, and over what time. Telling a hard truth may hurt someone now yet help them later; leaving a group frees the individual but weakens the group. Current AI characters — in games, in applications, eventually in companion robots — largely lack this. They produce fluent words without a structured account of how an action lands on the people and groups around them. The P/G framework supplies exactly that missing capability: an engine that, given any action, estimates its rippling consequences on individuals and on groups.

Scope boundary. The MVP is scoped to fully-observable, NPC-first worlds, where every action and consequence is inspectable and predictions can be checked. Partially-observable real-world human contexts are a deliberate future extension, not part of this specification.

2. Design philosophy: one small universal core, many thin layers

The architecture is deliberately split, so that the core stays small, stable, and universal while specialization lives in thin layers above it.

- A universal core. One shared mechanism every agent runs: the P/G dynamics — how an action changes individual property states (P) and relationship/group fields (G). The core provides a culture-independent output schema; interpretation of culturally variable meaning happens in the adapter, not the core (a sharper claim than calling the core itself “culture-neutral measurement”).
- Thin layers on top. For each life-form, culture, or individual, a thin interpretation/personality layer applies different weights, thresholds, retention, and cultural reading over the same core measurements. The core stays fixed; the layer specializes.

This is what lets one consequence engine serve many different characters, and lets the system grow by adding layers rather than redesigning the core.

3. Decision style: move-by-move, not end-to-end

Deliberation is inspired by AlphaZero. Rather than one end-to-end network computing all the way to a final verdict, the agent works move-by-move: (1) generate a few candidate actions; (2) estimate each action's local ΔP/ΔG consequence using the core, looking ahead only a bounded horizon via MCTS-style planning; (3) integrate relational value and affect; (4) choose the next move, then advance and repeat. This keeps per-move computation bounded and the system tractable and evolvable. The behavioral layer specified in this document is the consequence-estimation core in step (2); planning and choice are downstream consumers.

4. The two-layer separation (the central discipline)

The framework strictly separates two layers, and most design errors come from conflating them.

	Behavioral layer (this document)	Psychological layer (downstream, deferred)
Question	What are the consequences of this action?	Given who I am, how do I feel and what do I choose?
Output	agent-neutral ΔP / ΔG impact field	emotion, appraisal, decision
Contains	consequence estimation only	personality weights w, thresholds, emotion dynamics, choice
Personality	none — identical for every agent	all of it lives here

Affect, in the full framework, arises from effective P change (positive from a rise in relevant P, negative from a fall), with G determining whose P matters; personality, thresholds, and habituation are layers above this core. All of that is psychological-layer work and is out of scope for the behavioral-layer build (recorded in Part V).

5. Two claims, stated carefully

C1 — agent-neutral, not objective. The layer's outputs are model-estimated, agent-neutral consequences. Agent-neutral because no personality, emotion, or preference of the acting agent enters the computation. Not objective in the strong sense: every ΔP and ΔG is conditioned on the world model, the causal assumptions, the chosen P/G dimension definitions, the time horizon, and the calibration data. The word "objective" is not used. Agent-neutral is also not observer-neutral: the representation is fixed by the system designer and calibration protocol, and a different research community could choose a different P/G vocabulary. Neutrality is to the acting agent, not to all observers.

C2 — the output format is world-invariant; meaning is not. Every world produces the same impact-field schema over the same P-dimension and G-variable names. World-specific mechanics and culturally variable meanings (betrayal, honor, authority, shame) are translated into the shared P/G representation by the domain adapter; all semantic variation lives in the adapter, none in the core. The layer does not claim actions or emotions mean the same thing across worlds.

Part II — The representations: P, G, and relations

6. P — the individual state vector

P describes one individual over eight named dimensions. A P-state gives absolute values in [-1,+1]; a ΔP gives model-estimated change in [-1,+1]. Dimensions are always named fields, never positional arrays.

Dimension	Meaning	+1 anchor	-1 anchor
health	bodily condition and physical function	fully healthy	dead / destroyed
safety	exposure to imminent physical/material threat	fully secure	under lethal active threat
resources	material/economic holdings usable by the agent	abundant vs needs	total loss, destitute

Dimension	Meaning	+1 anchor	-1 anchor
autonomy	capacity to choose/act on accurate information	informed, unconstrained	fully coerced or deceived
identity	stability of self-concept and social roles	secure, affirmed	destroyed / erased
reputation	standing in the eyes of relevant others	maximally esteemed	maximally disgraced
capability	skills, knowledge, effective powers	at full capability	all capability lost
psychological	short-run affective/cognitive condition	flourishing	acute breakdown

Anchoring protocol. Annotators score ΔP by paired comparison against these anchors; absolute scoring is not used. ±1 is reserved for changes that move a dimension to its anchor.

Dimension-assignment rules. Several P-dimensions overlap (reputation vs psychological vs identity; safety vs health; autonomy vs capability). To keep M4 labels clean, assign a dimension to an item only where a distinct causal pathway in the trace supports it; if two dimensions would be driven by the same single pathway, assign the one the pathway most directly targets.

Event	Dimension(s)	Distinct pathway
deception / withheld truth	autonomy (-)	corrupts informed choice
public shame / humiliation	reputation (-) + psychological (-)	external standing AND internal state — two pathways
role / membership loss	identity (-)	self-concept / social role
injury realized vs imminent danger	health (-) vs safety (-)	damage done vs exposure to threat
teaching / training	capability (+)	skill or effective power gained
resource transfer	resources (±)	holdings change

P.psychological vs the psychological layer. P.psychological is the behavioral-layer scalar for the short-run cognitive/affective consequence of an action (a magnitude). The full emotion dynamics — appraisal, personality thresholds, time-derivative states such as worry — are a generative process in the psychological layer and are deferred. One is a consequence value; the other is a process. They do not overlap.

7. G — the group state vector

Group impact is a vector over three named variables; there is no scalar "group utility." A G-state gives absolute values; a ΔG gives change. (Note the dual use: member_welfare names both a state value and a delta — both in [-1,+1], but they are different quantities; implementers must not conflate the ranges.)

- member_welfare — the rollup of member-level P changes, computed by rule R1 (§15). It is derived, never directly authored by the projector.
- cohesion — emergent: internal bond strength, trust density, cooperative disposition of the group as a unit.
- continuity — emergent: the prospect that the group persists as a functioning unit over the horizon.

G-variable anchors (provisional, for direction scoring). cohesion: +1 = maximally bonded/unified, -1 = total fracture of bonds. continuity: +1 = certain to persist as a functioning unit, -1 = certain to dissolve. At MVP, G-deltas are scored on direction (sign) only; magnitude calibration of cohesion and continuity is deferred to the calibration session (§19).

cohesion and continuity are latent composites estimated by rule, not free outputs of M4 — their sign is derived from the relation graph and trace; their magnitude is deferred:

```
cohesion / continuity -- latent composites, rule-derived (MVP):
  cohesion(g)   := latent( trust_density, reciprocity, low_conflict,
                         shared_identity )
                  estimated by RULE from RelationGraph edge r-vectors among
                  members + trace events touching those edges. NOT free-
```

```
                authored by M4.
continuity(g) := latent( membership_stability, succession_redundancy,
                         low single-member dependence )
                estimated by RULE from GroupState membership + trace.
MVP: SIGN is rule-derived (an event that severs a high-trust member edge
     -> cohesion -). Proxy weights + latent aggregation (magnitude)
     DEFERRED to calibration, alongside lambda and w_P.
```

8. Relations — the relation graph

Relations between entities are carried in a graph (schema in §12). Edges hold a high-dimensional r-vector (trust, affection, dependence, authority, history valence); the old ally/adversary ±1 survives only as a sign field. Memberships carry two distinct quantities that must never be conflated: the structural weight m (how strongly an individual is structurally part of a group) and identification id_strength (how strongly they identify with it). Neither is the personality weight w of the acting agent — w belongs to the psychological layer and never appears in this graph.

9. The two sublayers

Internally the behavioral layer is two sublayers, kept apart for the same reason the two top-level layers are: debuggability. A model that jumps straight from an action to "good/bad" is opaque; an intermediate trace is checkable.

Causal-consequence sublayer
action + world facts → a RawConsequenceTrace: the step-by-step chain of what happens

↓

Impact-projection sublayer
trace → the ImpactField: named ΔP/ΔG with per-item uncertainty

The trace is the unit of debugging, annotation, and validation: every projected impact must point back into it.

Part III — Implementation specification

10. Data-model conventions

- Five frozen schemas: WorldState (§11), RelationGraph (§12), RawConsequenceTrace (§13), ImpactField (§14, the canonical output), GroupState (§16).
- Every payload carries schema_version and vocab_version. These are decoupled axes: adding a mechanism or fact_type bumps vocab_version; changing a schema's shape bumps schema_version. A module may never emit a token outside its declared vocab_version's vocabularies.
- All P-values and deltas live in [-1,+1]; all confidences and uncertainties in [0,1].
- Horizons are a closed set: short / mid / long. World-specific tick spans map to these three in adapter config, not in the core.
- The natural key of an impact item is target_type | target_id | dimension | horizon — target_type is included so a person id and a group id that share a string never collide. It is the unit of target-matching, dedup, rerun-stability alignment (§18), and derived_from references.
- A separate storage key prefixes the natural key with world_id | tick | actor_id | action_id for globally-unique persistence, logs, and training rows. The natural key matches within one field; the storage key is unique across the corpus.
- delta is the projected, pre-saturation change, bounded to [-1,+1] but not clipped against the target's current state. Mechanical saturation (next-state clipping) is the world-model's job at state-update, not M4's (see §14).

11. WorldState schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "world_id": "npc_village_01", "tick": 4821,
  "entities": [
    { "id": "agent_a", "type": "person",
      "p_state": { "health":0.8, "safety":0.6, "resources":0.4,
                   "autonomy":0.7, "identity":0.5, "reputation":0.2,
                   "capability":0.6, "psychological":0.5 } }
  ],
  "groups": ["family_1", "guild_2"],   // ids only; detail in GroupState
  "facts": [
    { "id":"f_001", "text":"agent_a possesses tool_3",
      "fact_type":"possession",        // from FACT_TYPE vocabulary
      "source":"adapter", "confidence":1.0 }
  ]
}

FACT_TYPE vocabulary (closed; vocab_version "1"):    <-- frozen here
  possession        entity holds a resource or object
  location          entity is at a place
  relationship      a directed tie between two entities exists
  membership        entity belongs to a group
  role              entity holds a role or status in a group/world
  capability_state  entity has a skill or effective power
  condition         entity's bodily / physical condition
  knowledge         entity knows or believes a proposition
  commitment        entity is bound by a promise or obligation
  threat_exposure   entity is exposed to an imminent threat (pairs P.safety)
  observed_action   an action was witnessed by an entity (pairs M2 observer)
  norm              a rule in force in this world or group
  [failure value] unmappable      // confidence <= 0.3; never silently dropped
NOTE: a fact_type describes a normalized world FACT (WorldState); a
  mechanism describes a causal TRANSITION step (RawConsequenceTrace).
  A fact_type is NOT a mechanism. e.g. fact_type 'condition' may be acted
  on by mechanism 'physical_harm_attempt'.
```

WorldState is supplied by the world model through M1; the behavioral layer reads it and never writes it.

12. RelationGraph schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "nodes": ["agent_a","agent_b","family_1","marriage_1"],
  "edges": [
    { "from":"agent_a", "to":"agent_b",
      "r_vector": { "trust":0.6, "affection":0.4, "dependence":0.1,
                    "authority":0.0, "history_valence":0.5 },
      "sign": 1 }                      // ally(+1)/adversary(-1): sign only
  ],
  "memberships": [
    { "agent":"agent_a", "group":"family_1",
      "m":0.9,                         // structural weight (calib deferred)
      "id_strength":0.8 }              // identification (calib deferred)
  ]
}
```

13. RawConsequenceTrace schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "action_id": "act_017",
  "steps": [
    { "step_id":"trace_001", "t":0,
      "event":"agent_a reveals secret s_4 to agent_b",
      "mechanism":"information_transfer",  // from MECHANISM vocabulary
      "affected":["agent_b"], "confidence":0.9 },
    { "step_id":"trace_002", "t":1,
      "event":"agent_b updates belief about agent_c",
      "mechanism":"belief_update",
      "affected":["agent_b","agent_c"],
      "confidence":0.7, "parent":"trace_001" }   // causal chain link
```

```
  ],
  "depth_limit": 4, "truncated": false
}
```

```
MECHANISM vocabulary (closed; vocab_version "1"):
  physical_harm_attempt   possession_transfer    information_transfer
  belief_update           commitment_change      membership_change
  status_change           support_provision      deception
  norm_violation_detection
  [failure value] unmappable      // emitted by M3 when no mechanism fits
```

- Every step names a mechanism from the closed vocabulary; M5 rejects traces with unknown mechanisms.
- Every id in affected must exist in the WorldState or GroupState registry; hallucinated entities cause step rejection.
- parent links form the causal chain; depth is bounded by depth_limit. Truncation is flagged, never silent — and a truncated branch means "not computed," not "no effect"; reviewers must not read a missing long-horizon item as a zero.
- norm_violation_detection is descriptive: it records that a world/group norm was violated, never a moral judgment about it. Evaluation belongs to the psychological layer — keeping this descriptive is part of the anti-smuggling discipline.

14. ImpactField — canonical output schema

A flat list of impact items. Each item is one (target, dimension, horizon) cell with its own delta and its own uncertainty — directly usable as a training/evaluation row.

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "actor_id": "agent_a", "action_id": "act_017",
  "impact_field": [
    { "target_id": "agent_b", "target_type": "P",   // "P" | "G"
      "dimension": "autonomy",       // named P-dim or G-variable
      "delta": 0.3, "horizon": "short",   // delta in [-1,+1]
      "uncertainty": 0.2,            // in [0,1], per item
      "evidence_ref": "trace_001" }, // -> a step_id (primary items)

    { "target_id": "agent_b", "target_type": "P",
      "dimension": "psychological", "delta": -0.5,
      "horizon": "short", "uncertainty": 0.2,
      "evidence_ref": "trace_001" },

    { "target_id": "friend_ab", "target_type": "G",
      "dimension": "cohesion", "delta": -0.2,
      "horizon": "long", "uncertainty": 0.8,
      "evidence_ref": "trace_002" },

    { "target_id": "family_1", "target_type": "G",   // M5-DERIVED item
      "dimension": "member_welfare", "delta": 0.0,    // NOT 0.15: uncalibrated
      "horizon": "short", "uncertainty": 1.0,
      "derived": true,
      "calibration_status": "structural_only",  // until lambda, w_P are set
      "numeric_scoring": false,                 // excluded from sign/magnitude
      "derived_from": ["P|agent_a|resources|short",
                       "P|agent_b|resources|short"] }
  ]
}
// natural key = target_type | target_id | dimension | horizon
// (used for matching, dedup, rerun alignment, and derived_from refs)
```

- Required per item: target_id, target_type, dimension, delta, horizon, uncertainty, and provenance.
- Provenance has two forms. A primary item (authored by M4 from one trace step) carries evidence_ref → a step_id. A derived item (member_welfare, authored by M5 from several member-P items) carries derived:true and derived_from → a list of contributing item natural keys, and omits evidence_ref. M5's dangling-reference check applies to primary items only.

- For target_type "G", a projector-authored dimension must be cohesion or continuity; member_welfare items are produced only by M5 under rule R1.
- member_welfare and other uncalibrated G items carry calibration_status:"structural_only" and numeric_scoring:false, with delta 0.0 / uncertainty 1.0 — never shown with a calibrated-looking number before λ and w_P are set.

delta semantics and saturation. This contract keeps M4 a reusable projector rather than a state-transition module:

```
delta semantics + saturation (resolves the boundary question):
  In ImpactField, `delta` is the PROJECTED, PRE-SATURATION change for a
  (target,dimension,horizon). It is bounded to [-1,+1] but is NOT clipped
  against the target's CURRENT state value.
  Mechanical saturation happens ONLY at state-update, owned by the WORLD-
  MODEL (the behavioral layer reads WorldState, never writes it):
     state_after    = clip( state_before + delta , -1, +1 )
     delta_realized = state_after - state_before
  delta_realized, p_before, p_after are WORLD-MODEL state-transition fields,
  NOT behavioral-layer / M4 fields.
  M4 MAY read current state as CAUSAL CONTEXT (e.g. 'victim is starving' ->
  larger psychological delta) but MUST NOT use it for boundary arithmetic.
  Principle: M4 captures causal context (via the trace); state-update
  applies mechanical saturation. Applies to P and to G (cohesion,
  continuity). member_welfare: derived by M5, then realized downstream.
```

Magnitude and uncertainty are two different band systems; benchmark rows use the uncertainty band:

```
Two band systems (previously conflated):
  delta_magnitude_band   low: 0<|delta|<=0.3  med: 0.3<..<=0.6  high: 0.6<..<=1.0
  uncertainty_band       low: 0<=u<=0.3       med: 0.3<u<=0.6   high: 0.6<u<=1.0
  Benchmark rows use the UNCERTAINTY band (magnitude is not scored at MVP;
  the magnitude band is defined for future use only).
```

Two keys serve two jobs — matching within a field, and global persistence:

```
Two keys (different jobs):
  natural key  target_type|target_id|dimension|horizon
               -> matching, dedup, rerun-alignment WITHIN one ImpactField
  storage key  world_id|tick|actor_id|action_id|target_type|target_id|
               dimension|horizon
               -> persistence, logs, and training data (globally unique)
```

15. Group aggregation rules R1–R3, and member_welfare scope

The early “use the minimal group containing all affected members” (union-G) rule is retired: the minimal union may not exist, may be meaningless, and erases subgroup relational change. Three rules replace it, enforced by M5.

- R1 — within-group dedup. For a group’s member_welfare, each member contributes exactly once, even if reached through several of that group’s subgroups. The rollup applies the §15b protective-hybrid form over the group’s own members’ ΔP (form frozen in §15b; only λ and w_P deferred to calibration, §19).
- R2 — per-group emergent variables. cohesion and continuity are computed independently for every explicitly tracked group, including subgroups. No union substitution.
- R3 — subgroup preservation / no parent summing. A tracked subgroup’s emergent items are reported separately and never merged into the parent; a parent’s welfare is computed from its full membership directly, never by summing subgroup welfares.

15a. member_welfare scope (resolves the dedup ambiguity). member_welfare is a per-group derived scalar. "Counted once" is scoped to a single group’s rollup (R1) and to the parent/subgroup chain (R3) — it is NOT a global per-person ledger. Two overlapping sibling groups each independently reflect their shared members. A person in family_1 and guild_2 therefore contributes to both groups’ welfare; omitting them from one would make that group’s welfare misrepresent its own membership. (This corrects the earlier B10 reading; see §18.)

15b. Rollup form (frozen) and constants (deferred). The functional form of the rollup is frozen now — a protective hybrid — while its constants are deferred to calibration, so M5 can be built as a parameterized function today:

```
member_welfare rollup -- FORM frozen, constants deferred (-> calib):
  W_g(h) = lambda * mean_m_i( S_i(h) ) + (1 - lambda) * min_i( S_i(h) )
```

```
    mean_m  m-weighted mean over group g's OWN members (R1: each once)
    min     protective term: severe harm to one member is not averaged away
    S_i(h)  scalar collapse of member i's affected dP at horizon h:
            S_i = sum_d  w_P[d] * dP_i[d]        (over affected dims d)
  DEFERRED to the dG calibration session (named, currently UNSET):
    lambda  in [0,1]        protective-mix weight
    w_P[d]  P-dimension welfare weights -- value-laden (is a death
            commensurable with a reputation loss?); NOT defaulted here
  => M5 builds this as a PARAMETERIZED function. Until lambda and w_P are
     set, member_welfare items are STRUCTURAL-ONLY (presence, R1, R3,
     consistency) and excluded from numeric sign/magnitude scoring.
```

What M5 gates at MVP: within-group dedup, no parent/subgroup summing, and the welfare↔member-P consistency rule. Rollup magnitude (λ, w_P) is deferred, so it is not gated.

16. GroupState schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "group_id": "family_1", "tracked": true,
  "members": [
    { "agent":"agent_a", "m":0.9, "id_strength":0.8 },
    { "agent":"agent_b", "m":0.9, "id_strength":0.7 }
  ],
  "subgroups": ["marriage_1"],         // explicitly tracked subgroups
  "g_state": {                         // STATE values (absolute), not deltas
    "member_welfare":0.55,             // rollup of member P; rule R1
    "cohesion":0.60,                   // emergent; calibration deferred
    "continuity":0.80 }                // emergent; calibration deferred
}
```

Only groups with tracked:true receive emergent-variable items; tracking is a world-model/adapter decision, not a behavioral-layer inference.

17. The five modules

The two sublayers are realized as five modules with hard JSON boundaries; every inter-module payload is validated against the schemas above.

M1 Domain Adapter — raw scene/action/world → normalized facts + relation-graph fragment

↓

M2 Candidate Target Detector — facts + action + graph → affected entities & groups (recall-first)

↓

M3 Raw Consequence Generator — action + facts + targets → RawConsequenceTrace

↓

M4 P/G Projector — trace → ImpactField (named dims, per-item uncertainty)

↓

M5 Aggregator / Validator — impact field + GroupStates → checked output (R1–R3, consistency)

M1 Domain Adapter. Output: normalized facts, a RelationGraph fragment, registry updates. Validation: referenced entities exist; fact_type and downstream tokens come from the closed vocabularies; no world-specific token (sword, spell, guild rank) in the output vocabulary — only in free-text event descriptions. Failure: an unmappable world mechanic becomes fact_type unmappable, confidence ≤ 0.3, never dropped. Test: a sword attack and a fire-spell attack both normalize to a physical_harm_attempt fact with identical schema (claim C2).

```
M1 world-invariance + provenance (worked):
  world A in:  "Knight swings sword at peasant"
  world B in:  "Mage casts firebolt at farmer"
  both -> fact { fact_type: condition, ... }        // target's body at risk
       -> (M3) mechanism: physical_harm_attempt
       -> identical ImpactField schema; only free-text 'event' differs (C2)
  M1 emits adapter_confidence per fact, SEPARATE from M3 step_confidence and
  M4 item uncertainty. Unmappable mechanic -> fact_type unmappable,
  adapter_confidence <= 0.3 (never silently dropped).
```

M1 strengthened contract. Mappings are defined by the adapter author in a reviewed per-world mapping table. Ambiguous cultural meaning is emitted as competing facts with lower adapter_confidence, never one guessed fact. Adapter quality is measured by fact-level precision/recall against a gold adapter set, plus unmappable_rate. Anti-smuggling rule (the key architectural guard): M1 output MUST NOT contain any ΔP/ΔG, affect, or normative/evaluative field — only facts and relations. Deltas are M4's job alone; a validation rule rejects any M1 payload carrying a delta or normative label. This is what stops the adapter from importing personality or moral judgment into the behavioral layer.

```
Vocabulary growth policy (vocab_version):
  v1 is INTENTIONALLY NARROW. New PRIMITIVE mechanisms are added via a
  versioned bump (v1 -> v2), never ad hoc. v2 shortlist (genuine primitives
  not expressible as compositions):
    threat / coercion      authority_command      access_restriction
  NOT primitives (compositions of v1 -- do NOT add):
    apology    = information_transfer + commitment_change
    forgiveness= commitment_change + status_change
    reward     = resource/status change (support_provision)
    punishment = resource/status change (norm_violation_detection branch)
  Coining a primitive per social act bloats the vocabulary and re-imports
  interpretation into the behavioral layer; resist it.
```

M2 Candidate Target Detector. Output: candidates with an inclusion_reason (direct_object, relation_neighbor, group_member, observer, norm_stakeholder). Validation: the actor is always included; each candidate is reachable within k hops or carries an observer/norm_stakeholder reason. Policy: recall-first — over-inclusion is fine (M4 may drop or zero them), omission is the failure that matters. Precision is recovered downstream. Bounded recall (avoids candidate blow-up in dense graphs): candidates are ranked by relation strength and capped (max_candidates, relation_strength_threshold, group_size_cap from config); pruned candidates are logged, not silently cut, so the policy stays recall-first rather than re-introducing precision filtering at M2. Acceptance checks that the pruned-expected-target rate is 0 on the benchmarks, so the caps can never silently drop a target that should have been found.

M3 Raw Consequence Generator. Output: RawConsequenceTrace. Validation: mechanisms from the vocabulary; affected ids registered; per-step confidence; acyclic parent links; depth ≤ depth_limit with explicit truncated flag. Failure: a step referencing an unregistered entity is rejected and logged; the trace is rebuilt without it, never silently repaired. Test: theft (B2) generates both the possession_transfer chain and the norm_violation_detection branch — traces are trees, not single chains.

```
M3 LLM execution contract (when a step uses an LLM):
  output           JSON-schema-validated against RawConsequenceTrace
  decoding         low temperature; fixed seed where available; constrained
                   decoding to the mechanism vocabulary
  retry            on schema-invalid output: <=2 retries, then fail the step
  bound            depth <= depth_limit; truncated flag set, never silent
  failure          a step that cannot be validated is rejected and logged
                   (M3 never emits an unvalidated step)
```

M4 P/G Projector. Output: ImpactField. Validation: deltas in range; dimensions from the schema names; primary items carry a resolving evidence_ref; projector-authored G items limited to cohesion/continuity. Uncertainty is authored by the composition heuristic below, not guessed. Failure — explicit ignorance: an unmapped mechanism projects to delta 0, uncertainty 1.0, never a confident guess. MVP method:

per-mechanism rules first; swap in a learned projector as labeled data accrues. G-item ownership: M4 may emit cohesion/continuity primary items only by applying declared rule templates over trace + RelationGraph + GroupState — never free-form judgment. It may read current state as causal context (e.g. a starving victim) but must not clip delta against it: delta is the projected, pre-saturation change (§14).

```
MVP uncertainty-composition heuristic (rule-based projector):
  uncertainty(item) = 1 - ( step_confidence
                          * rule_reliability[mechanism]
                          * horizon_factor[horizon] )
    horizon_factor:  short 1.00   mid 0.85   long 0.65
    rule_reliability: per-mechanism constant in the projector rule table
  -> longer horizons and lower-confidence steps yield higher uncertainty.
  Replaced by a learned calibrator once labeled data accumulates (§calib).
```

Emit-vs-drop. Two zero-delta items differ: "plausible but unknown" is kept at high uncertainty; "no pathway" is dropped. The deciding question is whether a trace step touches the target, not whether magnitude is known.

```
Emit-vs-drop rule for an item with delta ~ 0:
  EMIT  if a trace step causally touches the target (a pathway exists)
        but sign/magnitude is unknown -> delta 0 (or best-guess sign),
        uncertainty >= 0.7.   [ "plausible but unknown" ]
  DROP  if no trace step touches the target (M2 over-included, M3 found
        no pathway).          [ "no pathway -> no item" ]
A "spurious" item = a CONFIDENT (uncertainty < 0.5) non-zero item with
  no supporting trace step, or one contradicting an expected-absent label.
```

M5 Aggregator / Validator. Output: checked field, with member_welfare computed under R1. Validation: rules R1–R3; primary-item evidence_ref resolution; vocabulary/range checks; the welfare↔member-P consistency rule. Failure: a detected double-count or dangling reference rejects the field with a machine-readable diagnostic — never a silent repair. M5 is deterministic logic only, and is built first because it doubles as the test harness for every other module. Two modes: strict (test/train) rejects the whole field on any violation; runtime returns the valid items plus a diagnostics list, so an acting agent is not blocked when 90% of the field is sound. M5 is also the only module that derives member_welfare; M4 never authors it.

Part IV — Validation

18. Benchmark suite (10 scenarios)

Each scenario fixes a small cast, an action, and expected directional labels. Labels constrain direction and structure, not exact magnitude (bands: low ≤ 0.3 < med ≤ 0.6 < high). The suite doubles as the regression harness; every rule and failure mode is exercised by at least one scenario. The corrections below resolve the review's blocking issues: B3 now carries the member rows its welfare rolls up from; B10 now reflects per-group dedup correctly; B6/B9 list the previously-omitted antagonist/leader targets; expected-absent labels are stated where relevant.

B1–B10 are contract / regression tests, not evidence of generalization. They catch schema and rule failures and were designed around these rules; passing them is necessary, not sufficient. Generalization is measured only by the 100 held-out scenarios in §19, which are annotated independently and not used during design.

Metric definitions (apply to §19):

```
Metric definitions (make the acceptance test reproducible):
  cell match      emitted item matches an expected item iff it agrees on the
                  FULL natural key: target_type|target_id|dimension|horizon
  impact-cell     |matched expected cells| / |expected cells|   (primary)
   recall         (full-key match: right person AND dimension AND horizon)
  target-only     |matched target_ids| / |expected target_ids|  (secondary)
   recall         (right entity, even if dimension/horizon missed)
  sign accuracy   MACRO-averaged by dimension: per-dimension sign-correct
                  rate, then mean over dimensions (rare dims not drowned);
```

```
                 over determinate-sign items only; member_welfare excluded
  uncertainty    Spearman rank-corr( item uncertainty,
   calibration                       annotator sign-disagreement rate )
  annotators     >= 3 per scenario
  reliability    Krippendorff alpha >= 0.6 on sign; scenarios below the
                 floor are EXCLUDED from scoring (not counted as failures).
                 REPORT exclusion_rate; if > 0.2, the benchmark itself is
                 flagged (hard cases may be silently dropping out)
  sign disagree  expected sign = annotator majority; exact tie -> item is
                 indeterminate (?) and scored on uncertainty only
```

Grading rubric (applies to all ten):

```
Benchmark grading rubric:
  RECALL    every expected item's target must appear        (drives 18.1)
  SIGN      graded only on items with determinate (+/-) sign (drives 18.2)
            '?' and '~' items: excluded from sign; must carry unc >= 0.7
  HORIZON   graded to the named band (short/mid/long)
  ABSENT    an 'expected-absent' label is FAILED by any CONFIDENT
            (uncertainty < 0.5) item on that target/dimension
  EXTRA     additional high-uncertainty items are NOT penalized
            (consistent with recall-first M2)
  WELFARE   member_welfare rows are STRUCTURAL-ONLY: graded on presence/
            absence, R1 dedup, R3 non-summing, welfare<->member-P
            consistency; EXCLUDED from sign/magnitude until calibration
```

B1 — Painful truth. agent_a tells agent_b that b's late father caused the village fire. friend_ab is a tracked two-person group.

```
expected items (target | dim | sign | horizon | unc-band):
  agent_b | autonomy       | + | short | low
  agent_b | psychological  | - | short | low
  agent_b | psychological  | + | long  | high
  friend_ab | cohesion     | ? | long  | high   // ?-sign: score on unc only
expected-absent: no OTHER G items (only friend_ab may appear)
checks: per-item uncertainty spread; recall-first did not invent groups
```

B2 — Theft, observed. agent_a steals grain from agent_b; villager agent_c witnesses it. village_1 is tracked.

```
expected items:
  agent_a | resources      | + | short | low
  agent_b | resources      | - | short | low
  agent_b | psychological  | - | short | low
  agent_a | reputation     | - | mid   | med   // if c spreads word
  village_1 | cohesion     | - | mid   | med
  village_1 | member_welfare | (structural-only)        // sign deferred (calib)
checks: detection-risk branch present; welfare row tested for
        presence + consistency only, not sign
```

B3 — Group secret betrayed. agent_a (in guild_2) reveals guild_2's plan to rival guild_3 (members g3a, g3b).

```
expected items:                          // issue-1 fix: member rows added
  g3a | capability         | + | short | med   // a guild_3 member gains
  g3a | resources          | + | mid   | med
  guild_3 | member_welfare | (structural-only)   // supported by g3a rows
  guild_2 | cohesion       | - | short | low
  guild_2 | continuity     | - | mid   | med
  agent_a | reputation     | - | mid   | low   // within guild_2 context
checks: actor included; cross-group effects; welfare tested for support +
        presence only (sign deferred to calibration)
```

B4 — Gift. agent_a gives a tool to agent_b. Baseline positive scenario.

```
precondition: agent_b wants the tool; no hidden obligation,
  status asymmetry, or hostile context.
expected items:
  agent_a | resources      | - | short | low
  agent_b | resources      | + | short | low
  agent_b | psychological  | + | short | low
  friend_ab | cohesion     | + | short | med
```

```
expected-absent: NO negative items (gift is unambiguously prosocial)
checks: low-uncertainty baseline; no negative item hallucinated
```

B5 — Guild exit. agent_a publicly leaves guild_2. No member's individual P changes.

```
precondition: guild_2 is large enough that no individual
  member's resources, safety, or role changes.
expected items:
  agent_a | autonomy       | + | short | low
  agent_a | identity       | - | mid   | med
  guild_2 | cohesion       | - | short | low
  guild_2 | continuity     | - | mid   | med
expected-absent: NO guild_2 member_welfare item (no member P changed)
checks: consistency rule - welfare requires member-level P rows
```

B6 — Defending a member. agent_a physically defends member agent_b from outside attacker attacker_x.

```
expected items:                         // issue-8: attacker now listed
  agent_b | safety         | + | short | low
  agent_a | health         | - | short | med   // risk taken by defender
  attacker_x | capability  | - | short | med   // attack repelled
  guild_2 | cohesion       | + | short | low
  agent_a | reputation     | + | mid   | low
checks: actor's own risk included; antagonist target detected
```

B7 — White lie. agent_a falsely reassures agent_b that b's failing shop is fine.

```
expected items:
  agent_b | psychological  | + | short | low
  agent_b | autonomy       | - | short | low
  agent_b | resources      | - | mid   | high  // delayed correction cost
  friend_ab | cohesion     | ? | long  | high  // ?-sign: unc-only scoring
checks: short/long divergence as separate items
```

B8 — Affair revealed (subgroup test). agent_c reveals agent_a's affair; a is married to agent_b; marriage_1 ⊂ family_1; child agent_d.

```
expected items:                         // subgroup test (rules R2,R3)
  marriage_1 | cohesion    | - | short | low   // reported on the SUBGROUP
  marriage_1 | continuity  | - | mid   | med
  family_1 | cohesion      | - | short | low   // parent, as a unit
  agent_b | psychological  | - | short | low
  agent_d | psychological  | - | mid   | med   // child
  agent_a | reputation     | - | mid   | low
checks: R3 - marriage_1 items NOT merged into family_1; R1 - b,d once each
```

B9 — Public praise. leader agent_l praises agent_b before team_1; agent_e is a rival peer.

```
precondition: team_1 perceives the praise as accurate and fair.
expected items:
  agent_b | reputation     | + | short | low
  agent_b | psychological  | + | short | low
  agent_l | reputation     | + | short | low   // praising leader, mild
  team_1 | cohesion        | + | short | med
  agent_e | psychological  | - | short | high  // possible peer envy
checks: second-order target (peer e) detected by M2
```

B10 — Scarcity sharing (dedup test). agent_a shares food with agent_b in a shortage; both are in family_1 and guild_2.

```
expected items:                        // per-group dedup test
  agent_a | resources      | - | short | low
  agent_b | resources      | + | short | low
  agent_b | health         | + | short | low
  family_1 | member_welfare| (structural-only, unc high)  // sign deferred
  guild_2  | member_welfare| (structural-only, unc high)  // b NOT omitted
  family_1 | cohesion      | + | short | med
  guild_2  | cohesion      | + | short | high
checks: R1 - b counted ONCE PER GROUP, present in BOTH groups' welfare
        (sign excluded from scoring until lambda, w_P are calibrated)
```

19. MVP success criterion

```
MVP acceptance test (100 held-out NPC social scenarios):
  (1) impact-cell recall >= 0.90  full-key match; report target-only
                                 recall as a secondary metric
  (2) sign accuracy    >= 0.80   MACRO-by-dimension, determinate-sign items;
                                 member_welfare EXCLUDED (structural-only)
  (3) R1-R3 violations  = 0      validator-gated structural subset (below)
  (4) uncertainty calibration: Spearman(item uncertainty,
        annotator sign-disagreement) > 0.4
  (5) unmappable_rate:  0 on B1-B10 ;  < 0.15 on held-out scenarios
  (6) pruned-expected-target rate = 0 on B1-B10 (M2 caps must never drop
        an expected target); annotator exclusion_rate reported (flag >0.2)
  (7) beats LLM-only baseline on sign accuracy AND rerun stability
        (5 reruns; items aligned by natural key)
Annotation: >= 3 annotators/scenario; Krippendorff alpha >= 0.6 floor
  (scenarios below floor excluded, not failed); expected sign = majority,
  ties -> indeterminate (scored on uncertainty only).
R1-R3 GATED at MVP: per-group dedup, no parent/subgroup summing,
  welfare<->member-P consistency. Rollup MAGNITUDE (lambda, w_P) deferred
  to calibration, so not gated. Baseline = same model, schema in prompt,
  no M1-M5 decomposition, identical protocol.
```

Part V — Deferred work, downstream layer, and build order

20. Calibration session agenda (deferred)

The following are explicitly unresolved; the schemas above reserve their slots so none of them changes a module contract when answered.

- cohesion / continuity measurement — choose, define, and anchor proxies (interaction frequency, reciprocity, conflict/defection rate; membership churn, succession, single-member dependence).
- member_welfare rollup constants — the protective-hybrid form is frozen in §15b; calibration sets only λ and the P-dimension welfare weights w_P. (The form is no longer an open choice.)
- elicitation of m and id_strength — m from structural role, id_strength from behavioral identification; both kept strictly distinct from the personality weight w.
- the -1..+1 anchoring protocol for G-variables — paired-comparison annotation, agreement targets, adjudication.
- annotator handbook — one worked annotation per benchmark scenario, including explicit P.psychological examples that distinguish a short-run psychological consequence from a generated emotion (so annotators do not conflate the two).

21. Downstream psychological layer (out of scope, recorded)

These belong to the layer above the behavioral layer and are documented here only so the boundary is explicit. None of them is part of the current build.

- Emotion as P/G dynamics. Affect arises from effective P change (positive from a rise in relevant P, negative from a fall), with G selecting whose P matters; time-derivative terms give anticipatory states such as worry.
- Personality. Weights w, thresholds, retention/habituation, and cultural reading select which G-frame is active and how the same neutral impact field is experienced and acted on.
- Prosocial low-reactivity profile (formerly “AI perfect human”). A configuration that sets psychological-layer response thresholds toward maximum acceptance. It alters response thresholds, not the behavioral-layer consequence representation: anxiety-, shame-, and social-pain-relevant consequences are still fully computed in the impact field; risk signals and social-repair triggers are preserved.

- MCTS deliberation. Move-by-move planning that consumes the behavioral and psychological layers; not part of the core contribution.

22. Build order

- 1. Freeze the five schemas and the two vocabularies of this document.
- 2. Build M5 first: validator + a test harness loaded with B1–B10. Deterministic, cheap, and the gate for everything else.
- 3. Build M1 for one NPC world (the fully observable, NPC-first data regime).
- 4. Build M2, recall-first, evaluated on benchmark target lists.
- 5. Build M3 (rules + LLM) under M5's trace validation.
- 6. Build M4 as per-mechanism rules with the uncertainty heuristic; collect labeled rows from benchmark runs; swap in a learned projector when volume permits.
- 7. Iterate until B1–B10 pass under the §18 rubric.
- 8. Run the §19 acceptance test on 100 held-out scenarios against the LLM-only baseline.

MindOrigin / Noah — P/G 行为层

MVP 可执行规格说明书（v1.6.1）

状态：schema/合同冻结；标量聚合函数延后至校准 · 2026 年 6 月 · 自包含；取代 v1–v1.5 及统一设计记录

如何阅读本文档

这是一份单一的、自描述的文档。它把 P/G 框架的概念基础与其行为层的冻结实现规格合并在一起，并纳入了最近一次外部评审的全部修复。阅读它不需要任何先前版本。第一部分说明我们在做什么、为什么做。第二部分定义各项表示（P、G、关系）。第三部分是实现规格：数据模型、schema、模块与规则。第四部分是验证：基准测试集、评分与验收测试。第五部分记录延后事项与构建顺序。

标为绿色的段落是 v1.6 为响应最近一次评审而新增的内容——操作性规则（维度归属、adapter 合同、指标定义、member_welfare 形式），而非新哲学。v1.6.1 在其上加一层一致性补丁：渲染 FACT_TYPE 词表、修正 member_welfare 示例、定义 delta 饱和、分离幅度与不确定性 band、厘清 M4/M5 归属——无新理论。函数形式已冻结；其含价值判断的常数（λ、w_P）延后至 ΔG 校准 session，因此这是一份合同冻结的可执行规格，而非完全封闭的规格。代码块具有规范性：当散文与 schema 冲突时，以 schema 为准。

第一部分 — 基础：我们在做什么、为什么做

1. 问题

行动之前，人会下意识地估计一个行动的社会后果：影响到谁、以何种方式、有多大、跨多长时间。说出残酷真相，今天伤人、长远也许帮人；退出一个群体，个人更自由、群体却被削弱。今天的 AI 角色——游戏里、应用里、未来的陪伴机器人里——大多缺乏这种能力。它们能产出流畅的话语，却没有一个关于行动如何落在周围人和群体身上的结构化说明。P/G 框架恰好补上这块缺失的能力：一个引擎，给它任何一个行动，就估计出它对个人和群体的涟漪后果。

范围边界。MVP 限定于完全可观察的、NPC 优先的世界，其中每个行动与后果都可检视、预测可验证。部分可观察的真实世界人类情境是有意为之的未来扩展，不属于本规格。

2. 设计哲学：一个小而通用的核心，许多薄层

架构被刻意拆分，使核心保持小、稳定、通用，而专门化存在于其上的薄层中。

- 一个通用核心。每个 agent 都运行的同一个机制：P/G 动力学——一个行动如何改变个体属性状态（P）与关系/群体场（G）。核心提供一个文化无关的输出 schema；对文化可变意义的解释发生在 adapter 中，而非核心中（这比称核心本身为“文化中立的测量”更精确）。
- 其上的薄层。对每个生命形态、文化或个体，一个薄的解释/性格层在同一份核心测量之上施加不同的权重、阈值、保持度和文化解读。核心保持不变，薄层负责专门化。

正是这一点，让一个后果引擎能服务许多不同角色，并让系统通过增加薄层而非重新设计核心来成长。

3. 决策风格：逐步进行，而非端到端

深思借鉴了 AlphaZero。不是用一个端到端网络一路算到最终裁决，而是逐步进行：（1）生成少量候选行动；（2）用核心估计每个行动的局部 ΔP/ΔG 后果，只用 MCTS 式规划向前看一个有界的范围；（3）整合关系价值与情感；（4）选定下一步，然后推进并重复。这使每一步的计算有界，系统可处理且可演化。本文档规定的行为层，就是第（2）步中的后果估计核心；规划与选择是下游消费者。

4. 两层分离（核心纪律）

框架严格区分两个层，大多数设计错误都来自把它们混为一谈。

	行为层（本文档）	心理层（下游，延后）
问题	这个行动的后果是什么？	基于"我是谁"，我有什么感受、做什么选择？
输出	agent 中立的 ΔP / ΔG 影响场	情绪、评价、决策
包含	仅后果估计	性格权重 w、阈值、情绪动力学、选择
性格	无——对每个 agent 都相同	全部在这里

在完整框架中，情感源于有效的 P 变化（相关 P 上升产生正情感，下降产生负情感），由 G 决定谁的 P 重要；性格、阈值、习惯化是这个核心之上的层。这些全是心理层的工作，不在行为层构建范围内（记录于第五部分）。

5. 两个谨慎陈述的主张

C1 — agent 中立，而非客观。本层输出是模型估计的、agent 中立的后果。说 agent 中立，是因为行动主体的性格、情绪、偏好都不进入计算；说不是强意义上的客观，是因为每个 ΔP 和 ΔG 都依赖世界模型、因果假设、所选 P/G 维度定义、时间范围和校准数据。不使用"客观"一词。agent 中立也不是 observer 中立：表示由系统设计者与校准协议固定，另一个研究社区可能选择不同的 P/G 词表。中立是相对于行动主体的，而非相对于所有观察者。

C2 — 输出格式世界无关；意义不然。所有世界产出相同的影响场 schema，使用相同的 P 维度名与 G 变量名。世界特定机制与文化可变的意义（背叛、荣誉、权威、羞耻）由 domain adapter 翻译进共享的 P/G 表示；所有语义差异存在于 adapter 中，核心中没有。本层不主张行动或情绪在所有世界中意义相同。

第二部分 — 各项表示：P、G 与关系

6. P — 个体状态向量

P 用八个命名维度描述一个个体。P 状态给出 [-1,+1] 的绝对值；ΔP 给出 [-1,+1] 的模型估计变化。维度一律为命名字段，绝不用位置数组。

维度	含义	+1 锚点	-1 锚点
health（健康）	身体状况与生理机能	完全健康	死亡 / 被摧毁
safety（安全）	面临的即时人身/物质威胁程度	完全安全	处于致命现行威胁下
resources（资源）	可由个体支配的物质/经济持有	相对需求充裕	完全丧失、一无所有
autonomy（自主）	基于准确信息选择/行动的能力	信息充分、不受约束	完全被胁迫或欺骗
identity（身份）	自我概念与社会角色的稳定性	稳固、被肯定	被摧毁 / 抹除
reputation（声誉）	在相关他人眼中的地位	受最高尊崇	声名彻底败坏
capability（能力）	技能、知识与有效行动力	能力完整	能力全部丧失
psychological（心理）	短期情感/认知状况	心理繁盛	急性崩溃

锚定协议。标注者按与锚点的成对比较为 ΔP 打分，不使用绝对打分。±1 只用于把某维度推到锚点的变化。

维度归属规则。多个 P 维度存在重叠（reputation 对 psychological 对 identity；safety 对 health；autonomy 对 capability）。为保持 M4 标签干净，只在 trace 中存在独立因果路径支撑时才把某维度赋给一个条目；若两个维度由同一条路径驱动，赋给该路径最直接针对的那个。

事件	维度	独立路径
欺骗 / 隐瞒真相	autonomy (-)	腐蚀知情选择
当众羞辱	reputation (-) + psychological (-)	外部地位与内部状态——两条路径
失去角色 / 成员资格	identity (-)	自我概念 / 社会角色
已造成伤害 vs 迫近危险	health (-) vs safety (-)	已发生的损害 vs 面临威胁的暴露
教导 / 训练	capability (+)	获得技能或有效行动力
资源转移	resources (±)	持有变化

P.psychological 与心理层的区别。 P.psychological 是行为层中表示一个行动的短期认知/情感后果的标量（一个幅度）。完整的情绪动力学——评价、性格阈值、像“担忧”这类时间导数状态——是心理层中的生成过程，已延后。一个是后果值，另一个是过程。二者不重叠。

7. G — 群体状态向量

群体影响是三个命名变量上的向量；本层不存在标量"群体效用"。G 状态给出绝对值；ΔG 给出变化。（注意双重用法：member_welfare 既命名一个状态值，也命名一个 delta——二者都在 [-1,+1]，但是不同的量；实现者不得混淆其取值范围。）

- member_welfare（成员福祉） — 成员级 P 变化的汇总，由规则 R1 计算（§15）。它是派生量，绝不由投射器直接给出。
- cohesion（凝聚力） — 涌现变量：群体作为整体的内部纽带强度、信任密度、合作倾向。
- continuity（延续性） — 涌现变量：群体在该时间范围内作为有效运转单元持续存在的前景。

G 变量锚点（暂定，用于方向打分）。 cohesion：+1 = 最大限度地团结/凝聚，-1 = 纽带彻底破裂。continuity：+1 = 必定作为有效单元持续，-1 = 必定解体。在 MVP 阶段，G-delta 只按方向（符号）打分；cohesion 与 continuity 的幅度校准延后到校准 session（§20）。

cohesion 与 continuity 是按规则估计的潜在合成量，并非 M4 的自由输出——其符号由关系图与 trace 派生，其幅度延后：

```
cohesion / continuity -- latent composites, rule-derived (MVP):
  cohesion(g)   := latent( trust_density, reciprocity, low_conflict,
                           shared_identity )
                  estimated by RULE from RelationGraph edge r-vectors among
                  members + trace events touching those edges. NOT free-
                  authored by M4.
  continuity(g) := latent( membership_stability, succession_redundancy,
                           low single-member dependence )
                  estimated by RULE from GroupState membership + trace.
  MVP: SIGN is rule-derived (an event that severs a high-trust member edge
       -> cohesion -). Proxy weights + latent aggregation (magnitude)
       DEFERRED to calibration, alongside lambda and w_P.
```

8. 关系 — 关系图

实体间关系存在一张图里（schema 见 §12）。边持有高维 r-vector（信任、情感、依赖、权威、历史效价）；旧的盟友/敌对 ±1 只作为 sign 字段保留。memberships 携带两个必须严格区分的量：结构权重 m（个体在结构上属于某群体的强度）与认同强度 id_strength（其认同该群体的强度）。两者都不是行动主体的性格权重 w——w 属于心理层，绝不出现在这张图里。

9. 两个子层

行为层内部分为两个子层，分开的理由与两个顶层分开的理由相同：可调试性。一个从行动直接跳到"好/坏"的模型是不透明的；一个中间 trace 则可核查。

因果后果子层
行动 + 世界事实 → 一份 RawConsequenceTrace：事情发生的逐步链条

↓

影响投射子层
trace → ImpactField：命名的 ΔP/ΔG，逐条带 uncertainty

trace 是调试、标注与验证的基本单元：每一条投射出的影响都必须能回指到它。

第三部分 — 实现规格

10. 数据模型约定

- 五个冻结 schema：WorldState（§11）、RelationGraph（§12）、RawConsequenceTrace（§13）、ImpactField（§14，标准输出）、GroupState（§16）。
- 每个载荷携带 schema_version 与 vocab_version。这是两个解耦的轴：新增机制或 fact_type 时提升 vocab_version；改变 schema 形状时提升 schema_version。任何模块都不得输出其声明 vocab_version 词表之外的 token。
- 所有 P 值与 delta 在 [-1,+1]；所有 confidence 与 uncertainty 在 [0,1]。
- 时间范围是封闭集合：short / mid / long。各世界具体 tick 跨度在 adapter 配置中映射到这三档，不进入核心。
- 影响条目的自然键是 target_type | target_id | dimension | horizon——加入 target_type，使个人 id 与群体 id 即便字符串相同也绝不冲突。它是目标匹配、去重、重跑稳定性对齐（§18）与 derived_from 引用的单位。
- 另有一个存储键，在自然键前缀上 world_id | tick | actor_id | action_id，用于全局唯一的持久化、日志与训练行。自然键在单个 field 内匹配；存储键在整个语料中唯一。
- delta 是投射的、饱和前的变化，限定在 [-1,+1]，但不对目标当前状态裁剪。机械饱和（下一状态裁剪）是世界模型在状态更新时的工作，而非 M4 的（见§14）。

11. WorldState schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "world_id": "npc_village_01", "tick": 4821,
  "entities": [
    { "id": "agent_a", "type": "person",
      "p_state": { "health":0.8, "safety":0.6, "resources":0.4,
                   "autonomy":0.7, "identity":0.5, "reputation":0.2,
                   "capability":0.6, "psychological":0.5 } }
  ],
  "groups": ["family_1", "guild_2"],   // ids only; detail in GroupState
  "facts": [
    { "id":"f_001", "text":"agent_a possesses tool_3",
      "fact_type":"possession",         // from FACT_TYPE vocabulary
      "source":"adapter", "confidence":1.0 }
  ]
}
```

```
FACT_TYPE vocabulary (closed; vocab_version "1"):    <-- frozen here
  possession        entity holds a resource or object
  location          entity is at a place
  relationship      a directed tie between two entities exists
  membership        entity belongs to a group
  role              entity holds a role or status in a group/world
  capability_state  entity has a skill or effective power
  condition         entity's bodily / physical condition
  knowledge         entity knows or believes a proposition
  commitment        entity is bound by a promise or obligation
  threat_exposure   entity is exposed to an imminent threat (pairs P.safety)
```

```
  observed_action   an action was witnessed by an entity (pairs M2 observer)
  norm              a rule in force in this world or group
  [failure value] unmappable      // confidence <= 0.3; never silently dropped
NOTE: a fact_type describes a normalized world FACT (WorldState); a
  mechanism describes a causal TRANSITION step (RawConsequenceTrace).
  A fact_type is NOT a mechanism. e.g. fact_type 'condition' may be acted
  on by mechanism 'physical_harm_attempt'.
```

WorldState 由世界模型经 M1 提供；行为层只读不写。

12. RelationGraph schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "nodes": ["agent_a","agent_b","family_1","marriage_1"],
  "edges": [
    { "from":"agent_a", "to":"agent_b",
      "r_vector": { "trust":0.6, "affection":0.4, "dependence":0.1,
                    "authority":0.0, "history_valence":0.5 },
      "sign": 1 }                      // ally(+1)/adversary(-1): sign only
  ],
  "memberships": [
    { "agent":"agent_a", "group":"family_1",
      "m":0.9,                         // structural weight (calib deferred)
      "id_strength":0.8 }              // identification (calib deferred)
  ]
}
```

13. RawConsequenceTrace schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "action_id": "act_017",
  "steps": [
    { "step_id":"trace_001", "t":0,
      "event":"agent_a reveals secret s_4 to agent_b",
      "mechanism":"information_transfer",  // from MECHANISM vocabulary
      "affected":["agent_b"], "confidence":0.9 },
    { "step_id":"trace_002", "t":1,
      "event":"agent_b updates belief about agent_c",
      "mechanism":"belief_update",
      "affected":["agent_b","agent_c"],
      "confidence":0.7, "parent":"trace_001" }   // causal chain link
  ],
  "depth_limit": 4, "truncated": false
}

MECHANISM vocabulary (closed; vocab_version "1"):
  physical_harm_attempt   possession_transfer    information_transfer
  belief_update           commitment_change      membership_change
  status_change           support_provision      deception
  norm_violation_detection
  [failure value] unmappable      // emitted by M3 when no mechanism fits
```

- 每个 step 的 mechanism 取自封闭词表；含未知机制的 trace 由 M5 拒绝。
- affected 中的每个 id 必须存在于 WorldState 或 GroupState 注册表中；幻觉实体导致该 step 被拒绝。
- parent 链接构成因果链；深度受 depth_limit 约束。截断必须显式标记、绝不静默——被截断的分支意味着"未计算"，而非"无影响"；评审者不得把缺失的长期条目读作零。
- norm_violation_detection 是描述性的：它记录某条世界/群体规范被违反，绝非对其的道德判断。评价属于心理层——保持其描述性是防夹带纪律的一部分。

14. ImpactField — 标准输出 schema

一个扁平的影响条目列表。每条是一个（target, dimension, horizon）单元格，带有自己的 delta 与自己的 uncertainty——可直接作为训练/评估数据行。

```
{
```

```
  "schema_version": "1.0", "vocab_version": "1",
  "actor_id": "agent_a", "action_id": "act_017",
  "impact_field": [
    { "target_id": "agent_b", "target_type": "P",   // "P" | "G"
      "dimension": "autonomy",       // named P-dim or G-variable
      "delta": 0.3, "horizon": "short",   // delta in [-1,+1]
      "uncertainty": 0.2,            // in [0,1], per item
      "evidence_ref": "trace_001" }, // -> a step_id (primary items)

    { "target_id": "agent_b", "target_type": "P",
      "dimension": "psychological", "delta": -0.5,
      "horizon": "short", "uncertainty": 0.2,
      "evidence_ref": "trace_001" },

    { "target_id": "friend_ab", "target_type": "G",
      "dimension": "cohesion", "delta": -0.2,
      "horizon": "long", "uncertainty": 0.8,
      "evidence_ref": "trace_002" },

    { "target_id": "family_1", "target_type": "G",   // M5-DERIVED item
      "dimension": "member_welfare", "delta": 0.0,    // NOT 0.15: uncalibrated
      "horizon": "short", "uncertainty": 1.0,
      "derived": true,
      "calibration_status": "structural_only",  // until lambda, w_P are set
      "numeric_scoring": false,                 // excluded from sign/magnitude
      "derived_from": ["P|agent_a|resources|short",
                       "P|agent_b|resources|short"] }
  ]
}
// natural key = target_type | target_id | dimension | horizon
// (used for matching, dedup, rerun alignment, and derived_from refs)
```

- 每条必填：target_id、target_type、dimension、delta、horizon、uncertainty，以及来源（provenance）。
- 来源有两种形式。primary（主要）条目（由 M4 从一个 trace step 给出）携带 evidence_ref → 一个 step_id。derived（派生）条目（member_welfare，由 M5 从若干成员级 P 条目给出）携带 derived:true 与 derived_from → 贡献条目自然键的列表，并省略 evidence_ref。M5 的悬空引用检查只对 primary 条目生效。
- 当 target_type 为 "G" 时，投射器给出的 dimension 只能是 cohesion 或 continuity；member_welfare 条目只由 M5 按规则 R1 产生。
- member_welfare 及其他未校准的 G 条目携带 calibration_status:"structural_only" 与 numeric_scoring:false，delta 0.0 / uncertainty 1.0——在 λ 与 w_P 设定前，绝不以看似已校准的数字呈现。

delta 语义与饱和。 这条合同使 M4 保持为可复用的投射器，而非状态转移模块：

```
delta semantics + saturation (resolves the boundary question):
  In ImpactField, `delta` is the PROJECTED, PRE-SATURATION change for a
  (target,dimension,horizon). It is bounded to [-1,+1] but is NOT clipped
  against the target's CURRENT state value.
  Mechanical saturation happens ONLY at state-update, owned by the WORLD-
  MODEL (the behavioral layer reads WorldState, never writes it):
     state_after    = clip( state_before + delta , -1, +1 )
     delta_realized = state_after - state_before
  delta_realized, p_before, p_after are WORLD-MODEL state-transition fields,
  NOT behavioral-layer / M4 fields.
  M4 MAY read current state as CAUSAL CONTEXT (e.g. 'victim is starving' ->
  larger psychological delta) but MUST NOT use it for boundary arithmetic.
  Principle: M4 captures causal context (via the trace); state-update
  applies mechanical saturation. Applies to P and to G (cohesion,
  continuity). member_welfare: derived by M5, then realized downstream.
```

幅度与不确定性是两套不同的 band；基准行使用不确定性 band：

```
Two band systems (previously conflated):
  delta_magnitude_band   low: 0<|delta|<=0.3  med: 0.3<..<=0.6  high: 0.6<..<=1.0
  uncertainty_band       low: 0<=u<=0.3       med: 0.3<u<=0.6   high: 0.6<u<=1.0
  Benchmark rows use the UNCERTAINTY band (magnitude is not scored at MVP;
  the magnitude band is defined for future use only).
```

两个键服务两种用途——field 内匹配与全局持久化：

```
Two keys (different jobs):
  natural key  target_type|target_id|dimension|horizon
               -> matching, dedup, rerun-alignment WITHIN one ImpactField
  storage key  world_id|tick|actor_id|action_id|target_type|target_id|
               dimension|horizon
               -> persistence, logs, and training data (globally unique)
```

15. 群体聚合规则 R1–R3 及 member_welfare 范围

早期的"使用包含所有受影响成员的最小群体"（union-G）规则废除：最小并集可能不存在、可能无意义，并且会抹掉子群体的关系变化。改为以下三条规则，由 M5 执行。

- R1 — 群体内去重。对一个群体的 member_welfare，每个成员恰好贡献一次，即使经由该群体的多个子群体到达。汇总是该群体自身成员 ΔP 应用 §15b 的保护性混合形式（其常数 λ、w_P 延后至校准，§20）。
- R2 — 涌现变量按群体独立计算。cohesion 与 continuity 对每个显式追踪的群体（含子群体）独立计算。不允许并集替代。
- R3 — 子群体保留 / 不向父群体加总。被追踪子群体的涌现条目独立报告，绝不并入父群体；父群体的福祉直接由其完整成员计算，绝不通过加总子群体福祉得出。

15a. member_welfare 范围（解决去重歧义）。member_welfare 是按群体的派生标量。"只计一次"的范围限定在单个群体的汇总内（R1）以及父/子群体链内（R3）——它不是一个全局的按人台账。两个重叠的同级群体各自独立反映它们共享的成员。因此一个同时属于 family_1 和 guild_2 的人，对两个群体的福祉都有贡献；把他从其中一个里省略，会使那个群体的福祉错误地反映其自身成员。（这纠正了早先 B10 的读法；见 §18。）

15b. 汇总形式（冻结）与常数（延后）。汇总的函数形式现在冻结——一个保护性混合形式——而其常数延后至校准，因此 M5 今天即可作为参数化函数构建：

```
member_welfare rollup -- FORM frozen, constants deferred (-> calib):
  W_g(h) = lambda * mean_m_i( S_i(h) ) + (1 - lambda) * min_i( S_i(h) )
    mean_m  m-weighted mean over group g's OWN members (R1: each once)
    min     protective term: severe harm to one member is not averaged away
    S_i(h)  scalar collapse of member i's affected dP at horizon h:
            S_i = sum_d  w_P[d] * dP_i[d]        (over affected dims d)
  DEFERRED to the dG calibration session (named, currently UNSET):
    lambda  in [0,1]        protective-mix weight
    w_P[d]  P-dimension welfare weights -- value-laden (is a death
            commensurable with a reputation loss?); NOT defaulted here
  => M5 builds this as a PARAMETERIZED function. Until lambda and w_P are
     set, member_welfare items are STRUCTURAL-ONLY (presence, R1, R3,
     consistency) and excluded from numeric sign/magnitude scoring.
```

M5 在 MVP 阶段实际把关的：群体内去重、不向父/子群体加总、以及福祉↔成员级 P 的一致性规则。汇总幅度（λ、w_P）延后，故不把关。

16. GroupState schema

```
{
  "schema_version": "1.0", "vocab_version": "1",
  "group_id": "family_1", "tracked": true,
  "members": [
    { "agent":"agent_a", "m":0.9, "id_strength":0.8 },
    { "agent":"agent_b", "m":0.9, "id_strength":0.7 }
  ],
  "subgroups": ["marriage_1"],         // explicitly tracked subgroups
  "g_state": {                         // STATE values (absolute), not deltas
    "member_welfare":0.55,             // rollup of member P; rule R1
    "cohesion":0.60,                   // emergent; calibration deferred
    "continuity":0.80 }                // emergent; calibration deferred
}
```

只有 tracked:true 的群体才产生涌现变量条目；是否追踪是世界模型/adapter 的决定，不是行为层的推断。

17. 五个模块

两个子层实现为五个模块，模块间以硬性 JSON 边界衔接；所有模块间载荷都按上述 schema 验证。

M1 领域适配器 — 原始场景/行动/世界 → 标准化事实 + 关系图片段

↓

M2 候选目标检测器 — 事实 + 行动 + 关系图 → 受影响个体与群体（召回优先）

↓

M3 原始后果生成器 — 行动 + 事实 + 目标 → RawConsequenceTrace

↓

M4 P/G 投射器 — trace → ImpactField（命名维度，逐条 uncertainty）

↓

M5 聚合 / 验证器 — 影响场 + 群体状态 → 校验后输出（R1–R3，一致性）

M1 领域适配器。输出：标准化事实、RelationGraph
片段、注册表更新。验证：被引用实体存在；fact_type 及下游 token
取自封闭词表；输出词表中不出现世界特定
token（剑、法术、公会等级）——只在自由文本事件描述里出现。失败：无法映射的世界机制变为 fact_type unmappable、confidence ≤ 0.3，绝不丢弃。测试：挥剑攻击与火球术攻击都标准化为 schema 完全相同的 physical_harm_attempt 事实（主张 C2）。

```
M1 world-invariance + provenance (worked):
  world A in:  "Knight swings sword at peasant"
  world B in:  "Mage casts firebolt at farmer"
  both -> fact { fact_type: condition, ... }        // target's body at risk
       -> (M3) mechanism: physical_harm_attempt
       -> identical ImpactField schema; only free-text 'event' differs (C2)
  M1 emits adapter_confidence per fact, SEPARATE from M3 step_confidence and
  M4 item uncertainty. Unmappable mechanic -> fact_type unmappable,
  adapter_confidence <= 0.3 (never silently dropped).
```

M1 强化合同。映射由 adapter 作者在每个世界一份、经评审的映射表中定义。含糊的文化意义以多个较低 adapter_confidence 的竞争事实输出，绝不输出一个猜测的事实。adapter 质量以相对金标 adapter 集的事实级 precision/recall 加 unmappable_rate 衡量。防夹带规则（关键架构护栏）：M1 输出绝不得包含任何 ΔP/ΔG、情感或规范性/评价性字段——只有事实与关系。delta 是 M4 独有的工作；任何携带 delta 或规范性标签的 M1 载荷都被验证规则拒绝。正是这条阻止了 adapter 把性格或道德判断带入行为层。

```
Vocabulary growth policy (vocab_version):
  v1 is INTENTIONALLY NARROW. New PRIMITIVE mechanisms are added via a
  versioned bump (v1 -> v2), never ad hoc. v2 shortlist (genuine primitives
  not expressible as compositions):
    threat / coercion      authority_command      access_restriction
  NOT primitives (compositions of v1 -- do NOT add):
    apology     = information_transfer + commitment_change
    forgiveness= commitment_change + status_change
    reward      = resource/status change (support_provision)
    punishment = resource/status change (norm_violation_detection branch)
  Coining a primitive per social act bloats the vocabulary and re-imports
  interpretation into the behavioral layer; resist it.
```

M2 候选目标检测器。输出：带 inclusion_reason（direct_object、relation_neighbor、group_member、observer、norm_stakeholder）的候选。验证：行动者必含；每个候选在 k 跳内可达，或携带 observer/norm_stakeholder 理由。策略：召回优先——多包含没关系（M4 可丢弃或置零），漏掉才是真正的失败。精确率在下游恢复。有界召回（避免稠密图中候选爆炸）：候选按关系强度排序并封顶（配置中的 max_candidates、relation_strength_threshold、group_size_cap）；被剪枝的候选记录而非静默剪除，使策略保持召回优先，而非在 M2 重新引入精确率过滤。验收会检查基准上被剪枝的预期目标率为

0，使封顶绝不会静默丢掉本应找到的目标。

M3 原始后果生成器。输出：RawConsequenceTrace。验证：机制取自词表；affected 中 id 已注册；每 step 带 confidence；parent 链接无环；深度 ≤ depth_limit 并显式标记 truncated。失败：引用未注册实体的 step 被拒绝并记录；trace 在剔除该 step 后重建，绝不静默修补。测试：盗窃（B2）同时生成 possession_transfer 链与 norm_violation_detection 分支——trace 是树而非单链。

```
M3 LLM execution contract (when a step uses an LLM):
  output           JSON-schema-validated against RawConsequenceTrace
  decoding         low temperature; fixed seed where available; constrained
                   decoding to the mechanism vocabulary
  retry            on schema-invalid output: <=2 retries, then fail the step
  bound            depth <= depth_limit; truncated flag set, never silent
  failure          a step that cannot be validated is rejected and logged
                   (M3 never emits an unvalidated step)
```

M4 P/G 投射器。输出：ImpactField。验证：delta 在范围内；维度名取自 schema；primary 条目携带可解析的 evidence_ref；投射器给出的 G 条目限于 cohesion/continuity。uncertainty 由下面的组合启发式给出，而非猜测。 失败——显式无知：未映射机制投射为 delta 0、uncertainty 1.0，绝不给自信的猜测。MVP 方法：先用逐机制规则；随带标签数据积累换入学习型投射器。G 条目归属：M4 仅可通过对 trace + RelationGraph + GroupState 应用已声明的规则模板来产出 cohesion/continuity primary 条目——绝非自由判断。它可读取当前状态作因果情境（如挨饿的受害者），但不得据此裁剪 delta：delta 是投射的、饱和前的变化（§14）。

```
MVP uncertainty-composition heuristic (rule-based projector):
  uncertainty(item) = 1 - ( step_confidence
                          * rule_reliability[mechanism]
                          * horizon_factor[horizon] )
    horizon_factor:  short 1.00   mid 0.85   long 0.65
    rule_reliability: per-mechanism constant in the projector rule table
  -> longer horizons and lower-confidence steps yield higher uncertainty.
  Replaced by a learned calibrator once labeled data accumulates (§calib).
```

输出还是丢弃。两种零 delta 条目不同："貌似存在但未知"以高 uncertainty 保留；"无路径"则丢弃。判定问题是某 trace step 是否触及该目标，而非幅度是否已知。

```
Emit-vs-drop rule for an item with delta ~ 0:
  EMIT  if a trace step causally touches the target (a pathway exists)
        but sign/magnitude is unknown -> delta 0 (or best-guess sign),
        uncertainty >= 0.7.    [ "plausible but unknown" ]
  DROP  if no trace step touches the target (M2 over-included, M3 found
        no pathway).           [ "no pathway -> no item" ]
A "spurious" item = a CONFIDENT (uncertainty < 0.5) non-zero item with
  no supporting trace step, or one contradicting an expected-absent label.
```

M5 聚合 / 验证器。输出：校验后的影响场，其中 member_welfare 按 R1 计算。验证：规则 R1–R3；primary 条目 evidence_ref 解析；词表/范围检查；福祉↔成员级 P 一致性规则。失败：检测到重复计数或悬空引用时，整个影响场被拒绝并给出机器可读诊断——绝不静默修复。M5 是纯确定性逻辑，且第一个构建，因为它兼作其他所有模块的测试框架。两种模式：严格（测试/训练）一旦有违规就拒绝整个场；运行时返回有效条目加一份诊断列表，使行动中的 agent 在 90% 的场都正确时不被阻塞。M5 也是唯一派生 member_welfare 的模块；M4 从不撰写它。

第四部分 — 验证

18. 基准测试集（10 个场景）

每个场景固定一组小型角色、一个行动和预期方向标签。标签约束方向与结构，不约束精确幅度（档位：low ≤ 0.3 < med ≤ 0.6 < high）。该测试集兼作回归框架；每条规则、每个失败模式都至少被一个场景覆盖。下面的修正解决了评审的阻塞性问题：B3 现在带上了其福祉汇总所依据的成员行；B10 现在正确反映按群体去重；B6/B9 列出了此前被省略的对手/领导者目标；在相关处给出 expected-absent（预期缺失）标签。

B1–B10 是合同 / 回归测试，不是泛化证据。它们用于捕捉 schema 与规则失败，且本就围绕这些规则设计；通过它们是必要而非充分条件。泛化只由 §19 的 100 个保留场景衡量，那些场景独立标注、设计期间不使用。

指标定义（适用于 §19）：

```
Metric definitions (make the acceptance test reproducible):
  cell match      emitted item matches an expected item iff it agrees on the
                  FULL natural key: target_type|target_id|dimension|horizon
  impact-cell     |matched expected cells| / |expected cells|   (primary)
   recall         (full-key match: right person AND dimension AND horizon)
  target-only     |matched target_ids| / |expected target_ids|  (secondary)
   recall         (right entity, even if dimension/horizon missed)
  sign accuracy   MACRO-averaged by dimension: per-dimension sign-correct
                  rate, then mean over dimensions (rare dims not drowned);
                  over determinate-sign items only; member_welfare excluded
  uncertainty     Spearman rank-corr( item uncertainty,
   calibration                        annotator sign-disagreement rate )
  annotators      >= 3 per scenario
  reliability     Krippendorff alpha >= 0.6 on sign; scenarios below the
                  floor are EXCLUDED from scoring (not counted as failures).
                  REPORT exclusion_rate; if > 0.2, the benchmark itself is
                  flagged (hard cases may be silently dropping out)
  sign disagree   expected sign = annotator majority; exact tie -> item is
                  indeterminate (?) and scored on uncertainty only
```

评分细则（适用于全部十个）：

```
Benchmark grading rubric:
  RECALL    every expected item's target must appear        (drives 18.1)
  SIGN      graded only on items with determinate (+/-) sign (drives 18.2)
            '?' and '~' items: excluded from sign; must carry unc >= 0.7
  HORIZON   graded to the named band (short/mid/long)
  ABSENT    an 'expected-absent' label is FAILED by any CONFIDENT
            (uncertainty < 0.5) item on that target/dimension
  EXTRA     additional high-uncertainty items are NOT penalized
            (consistent with recall-first M2)
  WELFARE   member_welfare rows are STRUCTURAL-ONLY: graded on presence/
            absence, R1 dedup, R3 non-summing, welfare<->member-P
            consistency; EXCLUDED from sign/magnitude until calibration
```

B1 — 痛苦的真相。agent_a 告诉 agent_b：b 已故的父亲是村庄火灾的肇事者。friend_ab 是被追踪的二人群体。

```
expected items (target | dim | sign | horizon | unc-band):
  agent_b | autonomy      | + | short | low
  agent_b | psychological | - | short | low
  agent_b | psychological | + | long  | high
  friend_ab | cohesion    | ? | long  | high   // ?-sign: score on unc only
expected-absent: no OTHER G items (only friend_ab may appear)
checks: per-item uncertainty spread; recall-first did not invent groups
```

B2 — 被目击的盗窃。agent_a 偷走 agent_b 的粮食；村民 agent_c 目击。village_1 被追踪。

```
expected items:
  agent_a | resources     | + | short | low
  agent_b | resources     | - | short | low
  agent_b | psychological | - | short | low
  agent_a | reputation    | - | mid   | med   // if c spreads word
  village_1 | cohesion    | - | mid   | med
  village_1 | member_welfare | (structural-only)       // sign deferred (calib)
checks: detection-risk branch present; welfare row tested for
        presence + consistency only, not sign
```

B3 — 出卖群体机密。agent_a（guild_2 成员）把 guild_2 的计划泄露给敌对的 guild_3（成员 g3a、g3b）。

```
expected items:                          // issue-1 fix: member rows added
  g3a | capability         | + | short | med   // a guild_3 member gains
  g3a | resources          | + | mid   | med
  guild_3 | member_welfare | (structural-only)   // supported by g3a rows
  guild_2 | cohesion       | - | short | low
  guild_2 | continuity     | - | mid   | med
  agent_a | reputation     | - | mid   | low   // within guild_2 context
checks: actor included; cross-group effects; welfare tested for support +
        presence only (sign deferred to calibration)
```

B4 — 赠礼。agent_a 送给 agent_b 一件工具。正向基线场景。

```
precondition: agent_b wants the tool; no hidden obligation,
  status asymmetry, or hostile context.
expected items:
  agent_a | resources      | - | short | low
  agent_b | resources      | + | short | low
  agent_b | psychological  | + | short | low
  friend_ab | cohesion     | + | short | med
expected-absent: NO negative items (gift is unambiguously prosocial)
checks: low-uncertainty baseline; no negative item hallucinated
```

B5 — 退出公会。agent_a 公开退出 guild_2。所有成员个体 P 均无变化。

```
precondition: guild_2 is large enough that no individual
  member's resources, safety, or role changes.
expected items:
  agent_a | autonomy       | + | short | low
  agent_a | identity       | - | mid   | med
  guild_2 | cohesion       | - | short | low
  guild_2 | continuity     | - | mid   | med
expected-absent: NO guild_2 member_welfare item (no member P changed)
checks: consistency rule - welfare requires member-level P rows
```

B6 — 保护成员。agent_a 挺身保护成员 agent_b 免受外部攻击者 attacker_x 伤害。

```
expected items:                          // issue-8: attacker now listed
  agent_b | safety         | + | short | low
  agent_a | health         | - | short | med   // risk taken by defender
  attacker_x | capability  | - | short | med   // attack repelled
  guild_2 | cohesion       | + | short | low
  agent_a | reputation     | + | mid   | low
checks: actor's own risk included; antagonist target detected
```

B7 — 善意谎言。agent_a 谎称 agent_b 经营惨淡的店铺其实状况良好。

```
expected items:
  agent_b | psychological  | + | short | low
  agent_b | autonomy       | - | short | low
  agent_b | resources      | - | mid   | high  // delayed correction cost
  friend_ab | cohesion     | ? | long  | high  // ?-sign: unc-only scoring
checks: short/long divergence as separate items
```

B8 — 外遇被揭露（子群体测试）。agent_c 揭露 agent_a 的外遇；a 与 agent_b 已婚；marriage_1 ⊂ family_1；孩子 agent_d。

```
expected items:                          // subgroup test (rules R2,R3)
  marriage_1 | cohesion    | - | short | low   // reported on the SUBGROUP
  marriage_1 | continuity  | - | mid   | med
  family_1 | cohesion      | - | short | low   // parent, as a unit
  agent_b | psychological  | - | short | low
  agent_d | psychological  | - | mid   | med   // child
  agent_a | reputation     | - | mid   | low
checks: R3 - marriage_1 items NOT merged into family_1; R1 - b,d once each
```

B9 — 公开表扬。领导者 agent_l 当着 team_1 表扬 agent_b；agent_e 是竞争同伴。

```
precondition: team_1 perceives the praise as accurate and fair.
expected items:
  agent_b | reputation     | + | short | low
```

```
  agent_b | psychological  | + | short | low
  agent_l | reputation     | + | short | low   // praising leader, mild
  team_1 | cohesion        | + | short | med
  agent_e | psychological  | - | short | high  // possible peer envy
checks: second-order target (peer e) detected by M2
```

B10 — 匮乏时期的分享（去重测试）。粮食短缺中 agent_a 与 agent_b 分享食物；两人同属 family_1 与 guild_2。

```
expected items:                         // per-group dedup test
  agent_a | resources      | - | short | low
  agent_b | resources      | + | short | low
  agent_b | health         | + | short | low
  family_1 | member_welfare| (structural-only, unc high)  // sign deferred
  guild_2  | member_welfare| (structural-only, unc high)  // b NOT omitted
  family_1 | cohesion      | + | short | med
  guild_2  | cohesion      | + | short | high
checks: R1 - b counted ONCE PER GROUP, present in BOTH groups' welfare
        (sign excluded from scoring until lambda, w_P are calibrated)
```

19. MVP 成功标准

```
MVP acceptance test (100 held-out NPC social scenarios):
  (1) impact-cell recall >= 0.90  full-key match; report target-only
                                  recall as a secondary metric
  (2) sign accuracy    >= 0.80   MACRO-by-dimension, determinate-sign items;
                                 member_welfare EXCLUDED (structural-only)
  (3) R1-R3 violations  = 0      validator-gated structural subset (below)
  (4) uncertainty calibration: Spearman(item uncertainty,
        annotator sign-disagreement) > 0.4
  (5) unmappable_rate:  0 on B1-B10 ;  < 0.15 on held-out scenarios
  (6) pruned-expected-target rate = 0 on B1-B10 (M2 caps must never drop
        an expected target); annotator exclusion_rate reported (flag >0.2)
  (7) beats LLM-only baseline on sign accuracy AND rerun stability
        (5 reruns; items aligned by natural key)
Annotation: >= 3 annotators/scenario; Krippendorff alpha >= 0.6 floor
  (scenarios below floor excluded, not failed); expected sign = majority,
  ties -> indeterminate (scored on uncertainty only).
R1-R3 GATED at MVP: per-group dedup, no parent/subgroup summing,
  welfare<->member-P consistency. Rollup MAGNITUDE (lambda, w_P) deferred
  to calibration, so not gated. Baseline = same model, schema in prompt,
  no M1-M5 decomposition, identical protocol.
```

第五部分 — 延后工作、下游层与构建顺序

20. 校准 session 议程（延后）

以下问题显式未解决；上文 schema 已为它们预留字段位，因此任一问题得到解答时都不会改变模块合同。

- cohesion / continuity 的测量 —
  选定、定义并锚定代理（交互频率、互惠、冲突/背叛率；成员流失、继任、对单一成员的依赖）。
- member_welfare 汇总常数 — 保护性混合形式已在 §15b 冻结；校准只确定 λ 与 P 维度福祉权重 w_P。（形式不再是开放选择。）
- m 与 id_strength 的获取 — m 来自结构性角色，id_strength 来自行为性认同；两者都与性格权重 w 严格区分。
- G 变量的 -1..+1 锚定协议 — 成对比较标注、一致性目标、仲裁。
- 标注者手册 — 每个基准场景配一个完整标注示例，并包含明确的 P.psychological 示例，区分短期心理后果与生成的情绪（使标注者不致混淆二者）。

21. 下游心理层（不在范围内，仅记录）

以下属于行为层之上的层，在此记录只为使边界明确。它们都不属于当前构建。

- 作为 P/G 动力学的情绪。情感源于有效的 P 变化（相关 P 上升为正，下降为负），由 G 选择谁的 P 重要；时间导数项给出"担忧"这类预期状态。
- 性格。权重 w、阈值、保持/习惯化和文化解读，决定哪个 G 框架被激活，以及同一份中立影响场如何被体验和据以行动。
- 亲社会低反应配置（原名 "AI perfect human"）。一种把心理层响应阈值设向最大接受度的配置。它改变响应阈值，而非行为层的后果表示：与焦虑、羞耻、社会痛苦相关的后果仍在影响场中被完整计算；风险信号与社会修复触发被保留。
- MCTS 深思。消费行为层与心理层的逐步规划；不属于核心贡献。

22. 构建顺序

- 1. 冻结本文档的五个 schema 与两个词表。
- 2. 先构建 M5：验证器 + 装载 B1–B10 的测试框架。确定性、低成本，并且是其他一切的闸门。
- 3. 为一个 NPC 世界构建 M1（完全可观察的、NPC 优先的数据策略）。
- 4. 构建 M2，召回优先，用基准目标列表评估。
- 5. 在 M5 的 trace 验证之下构建 M3（规则 + LLM）。
- 6. 以逐机制规则加 uncertainty 启发式构建 M4；从基准运行积累带标签数据行；数量足够时换入学习型投射器。
- 7. 迭代直到 B1–B10 在 §18 细则下通过。
- 8. 在 100 个保留场景上执行 §19 验收测试，与 LLM-only 基线对比。